Auditing the Winner: Critic Disagreement Adds No Detectable Failure Signal Beyond the Selector on SWE-Agent Trajectories

Working draft v0.6 — SilentFail-Bench; two benchmarks, 50 repositories, pre-registered throughout

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Abstract

When an agent commits the best of n attempts under a critic, the committed trajectory is where the critic's confidence and its errors coincide. A common proposal is to audit it with a disjoint panel of critics and escalate on disagreement. We pre-registered and ran this test with executed gold on SWE-bench Verified (41 instances, 9 repositories) and SWE-bench Multilingual (114 instances, 41 repositories): best-of-4 selection by a reward model, a checkpoint-disjoint panel, and the official test harness. Silent failures — committed, agent-declared-complete trajectories that fail hidden tests — were 35% and 45% of the population. On the registered endpoint — the out-of-fold gain in AUROC from adding panel disagreement to the selector's own score — disagreement adds nothing detectable: pooled over 47 failures and 46 repositories, the three estimators that use every row give -0.03, -0.01 and $+0.01$, every interval includes zero, and the loosest pooled upper bound is $+0.12$. Disagreement alone ranks failures at AUROC 0.51 and 0.57, indistinguishable from the selecting critic's own score, and this held across 4k–16k contexts, pruned overlapping windows, seven reward models from 1B to 32B including code-trained ones, two re-selection regimes, and a single-family panel of LLM judges. Point estimates that cleared a structural baseline on Verified (windowed disagreement 0.69; patch size 0.67; one 1B critic 0.71) all fell to chance on 41 repositories: no configuration produced a replicable signal. An unselected-candidate control shows that with this selector the committed candidate fails no more often than one passed over, while the selector's own score ranks failures at 0.81 among unselected candidates and 0.61 among committed ones on Verified — the selection-conditional gap measured directly. Test–retest shows the critics are stable ($r \geq 0.98$) and content-driven: the failure is in what disagreement measures. We argue that audit signals must be evaluated on the population the selector commits, against the selector's own score, with executed gold; we report an upper bound on the residual value of disagreement; and we document the instrument failures that would otherwise have produced a positive result.

1 Introduction

Best-of-n selection under a learned critic is now a default in agent scaffolds: generate several trajectories, score each with a reward model, commit the argmax. Three claims about this are easy to conflate, and this paper keeps them apart. The *theory* is that argmax selection on a noisy score induces the optimizer's curse [46]: among candidates, the top-scored one is disproportionately the one whose flaws the critic cannot see, so the score carries less information about the committed candidate than about the pool. What we *observe* is one instance of that theory measured directly: on SWE-bench Verified the selecting critic ranks failures at 0.81 among the candidates it passed over and at 0.61 among the

ones it committed, on the same instances. What we do *not* show is that selection raised the failure rate of the committed candidate — with the selector we used, it did not (§4.1). A committed trajectory that fails its task while carrying a high score is a *silent failure*; they are a third to a half of what this agent committed, whether or not selection put them there.

The obvious safeguard is a second opinion — or several. If a panel of critics, disjoint from the selecting critic, disagrees sharply about a committed trajectory, that disagreement might carry information the selecting critic has exhausted. The proposal behind this work stated the hypothesis as a falsifiable, selection-conditional estimand and committed in advance to publishing the null.

This paper reports that result. The idea that correlated critics share blind spots is not ours: the correlated-judge literature predicts it [2, 13, 15, 22, 23], and best-of-n reward hacking is well characterized [11, 21, 61]. What had not been done is the measurement: put a disjoint audit panel on the population a critic actually committed, with executed gold, structural baselines and pre-registered bars, and ask what it adds over the selector. Our contributions are:

- **A selection-conditional evaluation protocol for audit signals** — selected population, selector baseline, executed gold, pre-registered bars — with its first two-benchmark instantiation on SWE-agent trajectories. Candidate-pool discrimination does not measure residual auditing value on the selected population [9, 31]; we evaluate on the selected population directly.
- **A replicated null with an estimator-robust upper bound.** On the proposal's registered endpoint — out-of-fold AUROC of a model with the selector's score and panel disagreement over one with the score alone — disagreement's incremental value is indistinguishable from zero under four estimators of it, with a loosest pooled upper bound of +0.12 over 47 positives and 46 repositories, and +0.25 on the smaller benchmark alone (§4.6). No single-protocol comparison in the sweep exceeds +0.22 at its upper interval, and the widest reaches +0.45; positives that appeared on twelve Verified failures did not travel.
- **Diagnostics and an instrument record:** test–retest stability of long-window reward-model scores; the repository-clustering ceiling on effective n for SWE-bench Verified; and six instrument failures that each produced a plausible false finding.

Throughout, "critic" means an off-the-shelf reward model or LLM judge used without task-specific training. Trained trajectory verifiers [39, 45, 51] are a different class; we return to them in §4.5.

2 Setup and estimand

Unit and gold. The unit is one committed trajectory τ. Gold $g(\tau) \in \{0, 1\}$ is trajectory-level task success under the official `swebench` 5.x harness; failed tests do not imply every step was wrong, and we make no step-level claims. Positive class throughout is failure (g=0).

Selection and audit. c_{sel} scores the full committed trajectory text; best-of-n ranks n full candidates by this score and commits the top-1. The audit panel $\{c_1, \ldots, c_m\}$ satisfies $c_{\text{sel}} \notin \{c_i\}$ at checkpoint level; u is their disagreement.

Primary population. Committed trajectories whose agent exited with a submission and a non-empty patch — the agent believed it was done. Within it, silent failures are g=0.

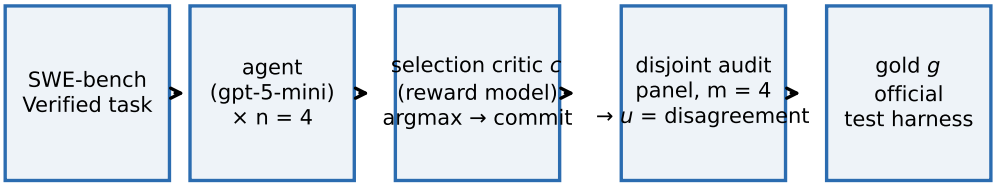


Figure 1: The design. c is the selecting critic's scalar score of a whole trajectory; best-of-n commits the argmax. A disjoint panel of m reward models produces u, the sample standard deviation of their rank-normalized scores (rank norms fit on training repositories only). Gold g is the official SWE-bench harness verdict on the committed patch.

Metrics. M1: AUROC(u) for silent failure. M2: AUROC($-c$). M3 = M1 − M2 with a repository-clustered bootstrap (4,000 resamples, groups resampled with the fitted scores held fixed). A held-out logistic ΔAUROC was pre-specified as confirmatory only when ≥ 10 positives fall in the test split; with 41 instances it never did, and we report raw-score AUROCs as registered.

Pre-registration. Population, positive class, metrics, predictions and a stopping rule ("fewer than ten positives yields no conclusion in either direction") were committed to the repository before any gold existed (`PREREGISTRATION_SWEBENCH_PILOT.md`). Each later analysis was added as a dated amendment before its data existed. Every deviation is declared in §6.

3 Instrument, and what it took to trust it

The pilot runs on a laptop and free GPU tiers: mini-swe-agent 2.4.6 driving `gpt-5-mini` through the OpenAI API ($0.02 per trajectory with prompt caching), SWE-bench images under Rosetta in a Colima VM on Apple Silicon, reward models on Metal or on Kaggle's free T4s, and the official harness for gold.

We record six instrument failures because each would have produced a publishable-looking result, and each was caught by an oracle control or a regression test rather than by inspection of the numbers.

1. A sandbox flag hid `numpy` from the gold harness: every trajectory scored as a failure. Caught because the benchmark's reference solutions are run through the harness before any model patch.
2. An LLM-as-judge selecting critic emitted three distinct values over 640 candidates, so best-of-16 silently became first-draw: 39 of 40 committed trajectories were identical to the n=4 run. Selection had not happened in the earlier pilots. Caught by the chosen-index histogram.
3. On a fresh VM the harness's image pulls failed on an architecture manifest: 41/41 failures in 499 seconds. Caught by the harness's own error count.
4. Our power function reported ≈0.40 "power" under the true null. Caught by simulating the null; replaced by a null-calibrated clustered bootstrap.

5. Repository grouping truncated hyphenated repository names, corrupting the held-out split. Caught by a regression test on the identifier parser.
6. Rank normalization was fit on all rows, leaking test rows into the panel statistic. Caught by a test that fits on the training split only. Every u reported in this paper is post-fix: recomputing the stored values from raw panel scores with training-repository normalization reproduces them exactly (max absolute difference 0.0 on both benchmarks), while the leaky computation differs by up to 0.023 and would have given $\mathrm{AUROC}(u) = 0.523$ rather than 0.508 on Verified.

The rule we adopted: a rate of exactly 0% or 100%, or a statistic that cannot move, is an instrument, not a result. The replication added two engineering stalls that produced no false finding (a per-image build layer and an API rate limit), recorded in the repository.

4 Results

4.1 Silent failures are common among committed trajectories

Fifty instances were sampled with a fixed seed; generation completed for 41 (the deviation is explained in §6). Of 164 candidates, 138 submitted a patch (17.7% hit the 40-step limit or submitted nothing). Selection by `Skywork-Reward-V2-Qwen3-4B` committed 34 usable trajectories; the harness resolved 22 and failed 12, with zero errors.

35.3% [21.5, 52.1] of committed, agent-declared-complete trajectories were silent failures on Verified, and 45.5% on Multilingual (§4.4). On HumanEval+ the same pipeline yielded 8% at any selection pressure; SWE-bench supplies the phenomenon in quantity. Selection was real (Fig. 2, top left), and $\mathrm{corr}(c, u) = 0.11$, so u was near-orthogonal to c — the precondition the design wants. Whether selection *concentrated* failure — whether the committed candidate fails more often than one the critic did not pick — is a separate question that a committed-only sample cannot answer; Amendment 4f, pre-registered before grading, answers it: one unselected candidate per instance was graded on both benchmarks (31 and 71 paired rows). **The committed candidate failed no more often than one the critic passed over**: -3 points on Verified $[-21, 0]$ and -4 on Multilingual $[-11, +3]$. With this selector, best-of-4 did not concentrate failure, and we claim a failure *rate*, not concentration. The control also measures the selection-conditional gap directly. On Verified the selector's own score ranks failures among the candidates it did *not* pick at 0.81 [0.74, 1.00], but among the ones it committed at 0.61 on the same instances: selection consumed the critic's own signal, which is the premise of the design. On Multilingual the critic has no signal in either arm (0.43 unselected, 0.51 committed), so there was nothing for selection to remove. Panel disagreement on the unselected arm is 0.37 and 0.51 — no better off-selection than on (Fig. 3).

4.2 The safeguard does not: the ladder of nulls

Table 1 and Fig. 4 are the core of the paper. The primary analysis — four chat-preference reward models auditing a 4,096-token prefix — gave $\mathrm{AUROC}(u) = 0.508$. Every subsequent amendment was declared before its data existed and asked whether some specific limitation of that primary was hiding the signal:

- *The critics never saw the patch.* Scoring the final diff plus the last 3,000 tokens: u fell to 0.458.
- *The critics were too small, or the wrong ones.* Six chat-preference RMs from 1B to 8B, including the proposal's named critic `Skywork-Reward-V2-Llama-3.1-8B`, each scored with the patch in view:

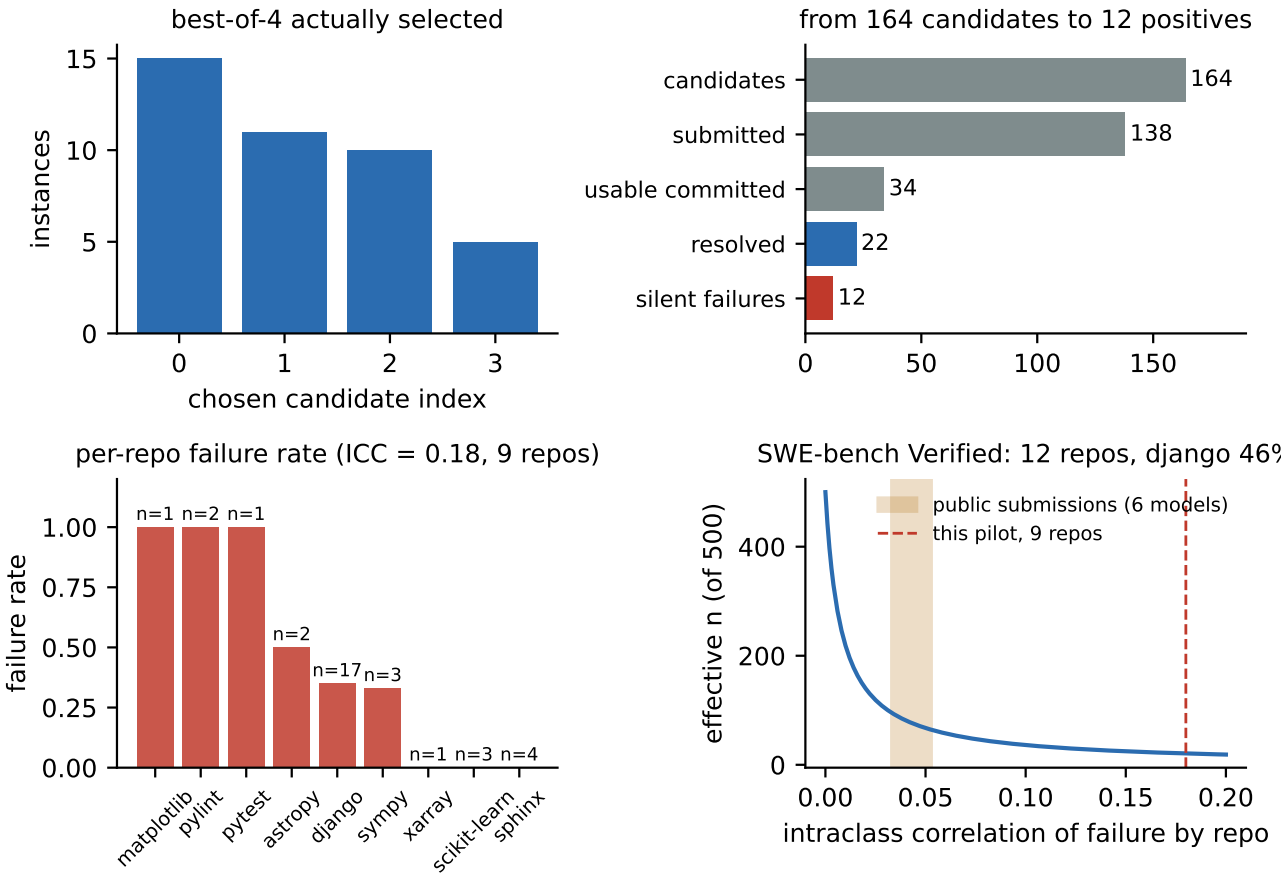


Figure 2: The pilot. Top left: best-of-4 with a true reward-model critic genuinely selected (chosen index spread across all four slots; an earlier LLM-judge critic had collapsed to index 0). Top right: from 164 candidates to 12 silent failures. Bottom left: per-repository failure rates, 9 repositories, ICC 0.18. Bottom right: effective sample size on SWE-bench Verified as a function of repository ICC; the public estimate from six leaderboard submissions is 0.033–0.053.

as individual detectors of the 12 known silent failures they scored 0.45–0.64, five of six at or below chance; the panel gave 0.494.

- *Selection was by the wrong critic.* Re-selecting with the named 8B critic changed 20 of 34 committed trajectories; graded, the failure rate was again 35%, and u was 0.445. Repeating the re-selection at a true 16k changed 23 of 33: u 0.483, while the critic's own score on the set it had itself selected reached 0.736.
- *Chat-preference training is the problem.* `AceCodeRM-7B`, trained on code with execution-feedback preferences, scored **0.405** as a detector — below chance — and adding it to the panel left u at 0.473. As a selector it changed 24 of 34 trajectories; u on that set was 0.542, its own score 0.472. Its 32B sibling, run in fp16 on an 80 GB GPU (131 of 164 rows at 8k, the rest at 16k), scored 0.557 as a detector and 0.50 by mean window.
- *The context was too short.* Re-scored on a 24 GB GPU at a true 16k tokens (every row at the declared length for the six chat RMs; the code-trained model fell back to 4k on 153/164 rows), five of six chat RMs improved as individual detectors — the 1B Skywork model to 0.712 — with paired 4k→16k gains of +0.06 to +0.12 whose intervals all include zero. The panel's disagreement at 16k

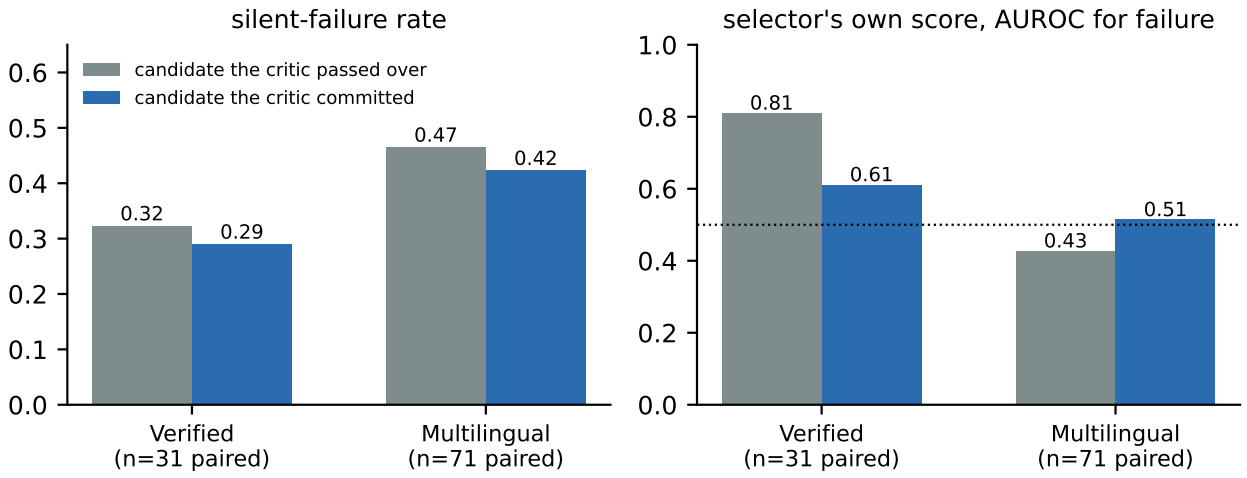


Figure 3: Amendment 4f. One candidate the critic passed over, graded per instance, paired with the one it committed. Left: selection did not raise the failure rate. Right: on Verified, selection removed the selector's own signal (0.81 → 0.61); on Multilingual the selector had none to remove.

was 0.439, the lowest of any row. An 8k point sits between and is not monotone.

- *Reward models are the wrong class.* Three LLM judges (`gpt-5`, `gpt-5-mini`, `gpt-5-nano`; one vendor, hence a single-family panel) rated the probability that the patch resolves the issue. Shown the problem and the diff, the two larger judges reached 0.72 and 0.75 — above the structural bar — but their disagreement was 0.528. Adding the last 6k tokens of the transcript, including the agent's own test runs, did not help (best judge −0.03; disagreement 0.640). `gpt-5-mini` is also the agent, a self-evaluation confound; `gpt-5` is not, and scored 0.72.

Every single-window prediction failed. The intervals on M3 are wide — twelve positives — and a moderate positive effect is not excluded by any one row; the rows are also re-measurements of the same twelve positives, not independent replications. What they establish jointly is narrower than "no detector": under every variation of input length, size, identity, training signal, critic class and selection regime, *the disagreement of a panel on one window of a committed trajectory* carries no information about its hidden failure that the selecting critic's own score does not.

4.3 Windowed auditing: the one row that clears the bar

Amendment 3, pre-registered before any windowed score existed, made one structural change: collapse tool outputs longer than 24 lines to their first and last 8, then score overlapping 3,500-token windows (25% overlap, problem statement prefixed, final window anchored at the patch) and take the panel's maximum disagreement. All three registered predictions passed: AUROC($u_{\max}$) = 0.686 against a bar of 0.665; a gain of +0.18 over the prefix panel against a bar of +0.10; and the named 8B critic's mean-window score, 0.652, against 0.634.

We report this as the study's strongest positive signal and immediately bound it. A free permutation of labels gives $p = 0.04$; permuting within repository, which removes the contribution of repositories that both fail more and disagree more, gives $p = 0.21$. Leaving out any one of the 34 instances moves the estimate between 0.657 and 0.731 — across the bar. The paired gain over the selecting critic is +0.06 [−0.11, +0.33]. Pruning removed 69% of tokens, so most trajectories fit in one or two windows

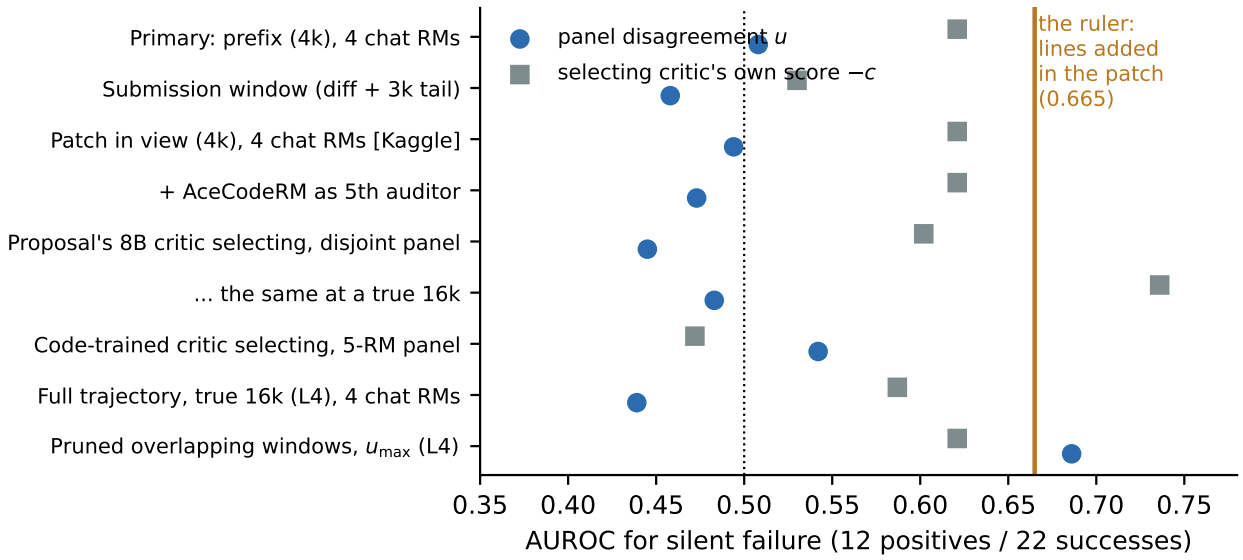


Figure 4: Every pre-registered configuration on the same 34 trajectories and 12 positives. Circles: panel disagreement u. Squares: the selecting critic's own score. Nothing crosses the ruler.

(median two); "multi-window" here mostly means "the pruned trajectory, plus one overlapping cut". In 43% of multi-window rows the maximum fell on a window other than the last, so the middle of the trajectory contributes something. $u_{\max}$ is also a maximum of noisy statistics and is biased upward under the null. A reader who takes 0.686 as a detector is reading more than twelve positives can support; a reader who dismisses it is ignoring the only pre-registered prediction in the study that came true.

4.4 Replication on 41 repositories: the null survives, the exceptions do not

The binding limitation of everything above is twelve repositories. Amendment 4e, pre-registered before any trajectory existed, re-ran the instrument on SWE-bench Multilingual: 41 repositories in nine languages, up to three instances each (114), k=4 (456 trajectories), the same selection critic and panel, agent sandboxes and gold on Modal (three reference-patch / empty-patch oracle pairs passed before any model patch was graded). Of 114 committed trajectories, 77 formed the population (30 empty patches, 7 ungradable); **35 were silent failures (45.5%)** across 37 repositories — three times the positives of the pilot, with an inference floor of 20 pre-registered.

The three registered predictions resolved as follows. **R-P1** (the null replicates): panel u on the 4k prefix, 0.565 [0.43, 0.71], against the critic's own 0.560; paired difference +0.005 [−0.19, +0.22]; within-repository permutation $p = 0.85$. *Passed.* **R-P2** (the windowed positive confirms): $u_{\max}$ 0.541 [0.40, 0.68], within-repository $p = 0.53$; leave-one-instance-out range 0.528–0.557 — the interval that on Verified straddled the bar now sits tightly on chance. *Failed.* **R-P3** is vacuous: $u_{\max}$ does exceed the lines-added baseline, because the baseline itself fell to 0.508 [0.35, 0.65]. Patch size predicted silent failure on Verified and does not here; it was a regularity of that benchmark, not of agents.

Table 2 and Fig. 6 carry the point. Nothing that rose above the ruler on Verified rose above chance on Multilingual: the best reward model at 16k is 0.577 (the 1B model that scored 0.712 is at 0.485); the named 8B critic 0.52; the code-trained model 0.356; every windowed single-critic aggregate 0.45–0.64. With 35 positives the intervals are half the width they were, and they now exclude the effects the

Configuration	AUROC(u)	AUROC($-c$)	M3	95% CI
Primary: 4k prefix, 4 chat RMs	0.508	0.621	−0.114	[−0.37, +0.22]
robustness, m=3 family-disjoint	0.545	0.621	−0.076	[−0.39, +0.23]
Submission window: diff + last 3k tok	0.458	0.530	−0.072	[−0.35, +0.13]
Patch in view (4k), 4 chat RMs	0.494	0.621	−0.127	[−0.28, +0.11]
+ AceCodeRM-7B as 5th auditor	0.473	0.621	−0.148	[−0.30, +0.10]
8B named critic selecting; disjoint panel	0.445	0.602	−0.157	[−0.56, +0.14]
the same at a true 16k (33 rows)	0.483	0.736	−0.252	[−0.47, +0.28]
Code-trained critic selecting; 5-RM panel	0.542	0.472	+0.069	[−0.21, +0.45]
Full trajectory, true 16k, 4 chat RMs	0.439	0.587	−0.182	[−0.38, +0.17]
LLM judges (gpt-5 family), patch in view	0.528	0.621	−0.093	[−0.30, +0.10]
execution-aware (+ last 6k tok)	0.640	0.621	+0.019	[−0.28, +0.16]
Pruned overlapping windows, $u_{\max}$	**0.686**	0.621	+0.064	[−0.11, +0.33]

Table 1: The pre-registered primary and all amendments. Columns are M1, M2 and M3 = M1 − M2 (§2). Same 34 trajectories; 12 positives in every row except the two re-selections (12 and 11; the 16k re-selection excludes one ungradable patch). M2 is the selecting critic's own score at the row's input where it was re-scored, otherwise the primary c. CIs are repository-clustered bootstraps on the paired difference. Only the last row clears the structural bar of 0.665 (§4.3).

Quantity on SWE-bench Multilingual 77 rows, 35 positives, 37 repositories	AUROC	95% CI / p
panel u, 4k prefix (primary)	0.565	$[0.43, 0.71]$
paired vs the selecting critic	+0.005	$[-0.19, +0.22]$
selecting critic $-c$	0.560	—
$u_{\max}$ over pruned windows	0.541	$[0.40, 0.68]$, $p_{\text{repo}} = 0.53$
u_{mean} / u_{last}	0.541 / 0.552	—
lines added (ruler)	0.508	$[0.35, 0.65]$
best single RM at 16k (Skywork-1.7B)	0.577	seven RMs: 0.36–0.58
named 8B critic, mean window	0.465	—
LLM judges: gpt-5 / mini / nano	0.662 / 0.604 / 0.576	each within its CI of c
LLM-judge disagreement u_{LLM}	0.514	$[-0.30, +0.10]$ vs c
registered endpoint, $M_{c+u} - M_c$ (LORO)	−0.022	$[-0.13, +0.08]$

Table 2: Every pre-registered and secondary quantity on the replication.

pilot's point estimates suggested.

4.5 What the critics do and do not carry

On Verified, single critics given enough of the solution reached 0.71–0.75 (the 1B reward model at 16k; the two larger LLM judges with the patch in view), while every disagreement statistic built on the same scores landed at 0.44–0.64. On Multilingual the reward-model half of that observation did not survive (§4.4), and the LLM-judge half shrank: with the patch in view, `gpt-5` 0.662, `gpt-5-mini` 0.604, `gpt-5-nano` 0.576 — each within its interval of the selecting critic — and their disagreement 0.514. The strongest single detector in the whole study is a frontier judge reading the diff, at two-thirds; nothing built from disagreement is above chance on either benchmark. Whatever the critics carry, disagreement is not extracting it.

Two things we did not test bound the claim from above. First, the selector is a 4B chat-preference reward model, not a deployed-grade trajectory verifier; whether a stronger selector leaves a different residual for an auditor is open, and Shum et al. [45] provide the natural instrument. Second, *trained*

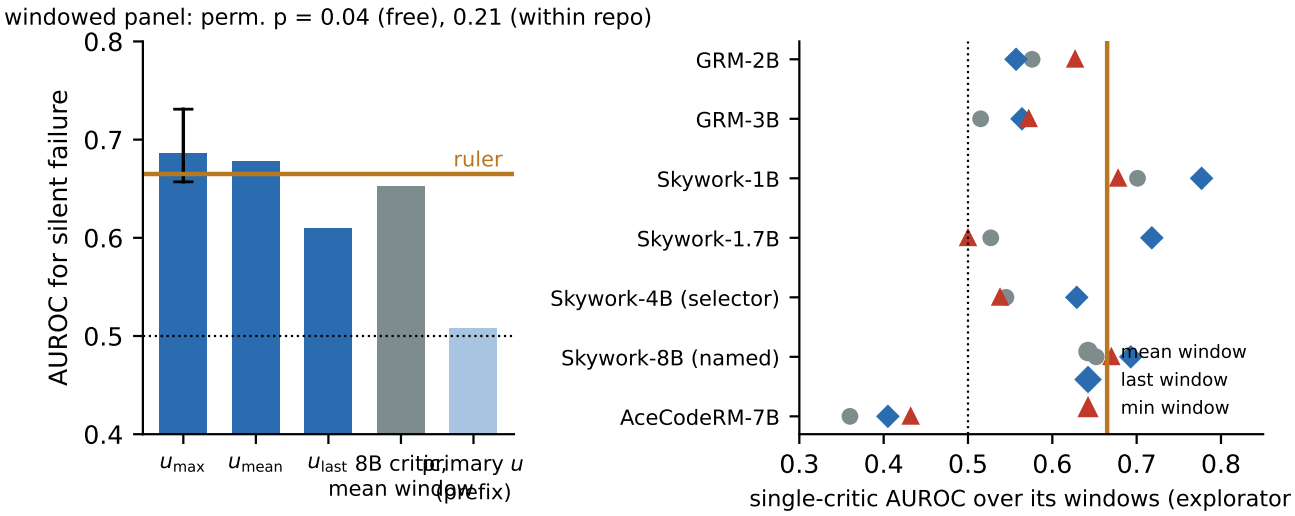


Figure 5: Amendment 3. Left: panel disagreement aggregated over pruned, overlapping windows; whisker on $u_{\max}$ is the leave-one-instance-out range. Right: each critic's own score aggregated over its windows (exploratory; no prediction was registered for these).

auditors exist and report better numbers on related populations: a 4B verifier trained on labelled failures selects among candidate trajectories and outperforms frontier judges at finding them [39]; a 0.6B prefix monitor predicts SWE-agent failure at a 5% false-positive rate [51]; lightweight statistical detectors beat LLM judges on agent-claimed false success [1]. Our result is about what a panel of *off-the-shelf* critics contributes through disagreement, evaluated the way an escalation trigger would be used; it says nothing against training an auditor, and the protocol here is how such an auditor should be evaluated. Second, a pre-registered incremental-value test (Amendment 4a) asked whether u or c adds anything to a structural model of the diff and transcript under leave-one-repository-out evaluation: u adds -0.03 $[-0.08, +0.04]$; c adds $+0.02$ $[-0.10, +0.21]$. We declare one defect in that amendment: the registered "difficulty" control, the fraction of six public leaderboard runs that solved the instance, turned out to predict our silent failures at 0.95 on its own — one of those runs is our own agent model, and the others fail the same instances — so the registered model was at ceiling and uninformative. The numbers above are from the labelled structure-only sensitivity variant; all three variants are in the record.

4.6 The registered endpoint: does u add to c out of fold?

The proposal's confirmatory quantity was the held-out ΔAUROC of a logistic model M_{c+u} over M_c. The 70/30 repository split never held ten test positives, so it was never eligible on Verified alone. With 47 positives across both benchmarks it can be computed by leave-one-repository-out, which we declare as a substitution of the cross-validation scheme, not of the estimand.

A two-parameter model refit on eight to thirty-six clusters is a fragile thing to hang a bound on, so we compute the endpoint four ways (Table 3): pooled out-of-fold probabilities from a leave-one-repository-out logistic; the same with five clustered folds; AUROC computed *inside* each fold and averaged by fold size, which avoids pooling probabilities across folds with different calibration; and a fixed unit-weight sum of the two standardized scores, which fits no coefficients at all. The four disagree in sign — pooled, they give -0.119, -0.030, -0.014 and $+0.013$ — and we therefore do

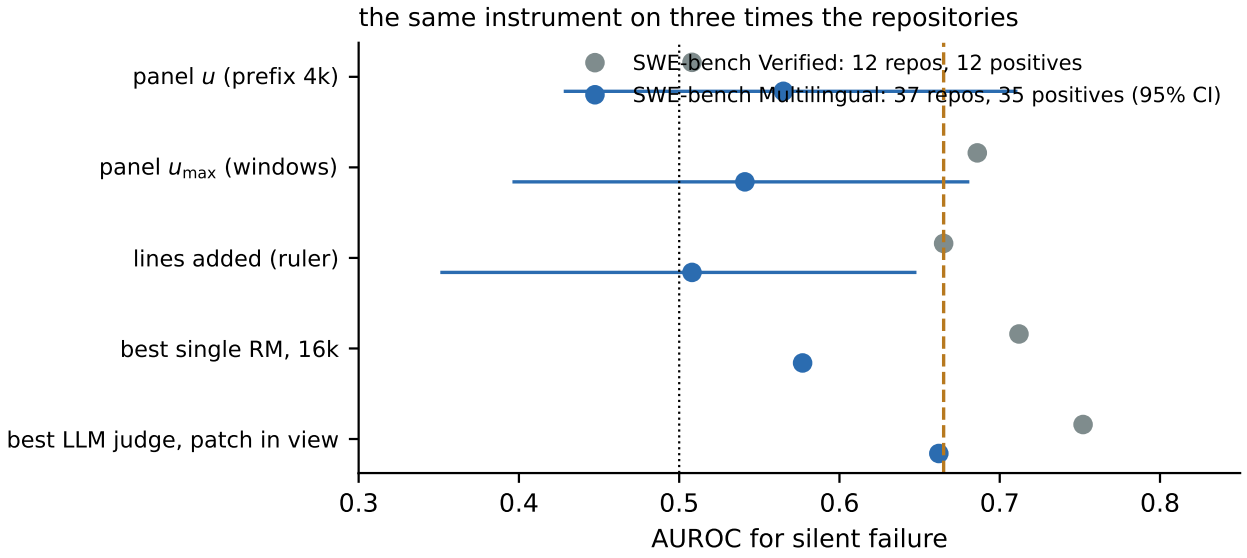


Figure 6: Amendment 4e. Every quantity that looked like a signal on Verified, re-measured on SWE-bench Multilingual with the same agent, critic, panel, pruning, windows and executed gold. Intervals are repository-clustered bootstraps.

not report any one of them as the estimate. What they agree on is the conclusion: every interval includes zero. The bound needs stating carefully. Among the three estimators that use every row, **the loosest pooled upper bound is** +0.12; per benchmark it is +0.25 (Verified, fixed weights), where twelve positives buy little precision. The fourth estimator, which computes AUROC inside each held-out repository, is the weakest: a single repository is usually all-pass or all-fail, so no AUROC is defined there and the fold is dropped — 15 of 46 folds and 53 of 111 rows survive, and its interval, $[-0.31, +0.28]$ pooled, excludes almost nothing. We report it for completeness and do not lean on it. The defensible statement is that adding disagreement to the selector's score produces no detectable gain under any of the four estimators we tested, with a pooled upper bound of +0.12 from those that use all the data.

The spread itself is instructive. The fixed-weight variant reproduces the raw AUROC$(-c)$ almost exactly (0.559 vs. 0.560 pooled), while the logistic variants report M_c below it (0.48–0.50): pooling out-of-fold probabilities across folds whose intercepts differ scrambles the global ranking, which depresses both arms roughly equally and cancels in the difference. Readers who prefer an assumption-free statement should take the fixed-weight row: +0.013 [−0.07, +0.12] pooled. The discrepancy is worth one sentence beyond this paper: with rank metrics and few clusters, pooling out-of-fold predictions can make a real signal appear to vanish when what has vanished is comparability across refits, and an unfitted combination or within-fold evaluation is the cheap check.

4.7 A ruler beats the critics — on one benchmark

Two integers, read off the diff and the transcript with no model at all, rank silent failures better than every reward model at 4k and every disagreement statistic at any length (Fig. 7): lines added in the patch, 0.665; agent step count, 0.657. Bigger patches and longer trajectories fail more. The paired gap over the primary u is +0.16 [−0.11, +0.36], exploratory, on twelve positives. Individual critics do cross this bar once they see the solution at length (§4.3); the panel's disagreement never does on

Estimator of the registered endpoint	Population	AUROC base	AUROC $+u$	ΔAUROC	95% CI
Logistic, leave-one-repo-out	Verified	0.527	0.451	-0.076	$[-0.16, +0.08]$
(pooled OOF probabilities)	Multilingual	0.459	0.436	-0.022	$[-0.13, +0.08]$
	Pooled	0.484	0.454	-0.030	$[-0.10, +0.04]$
Logistic, 5 clustered folds	Verified	0.644	0.455	-0.189	$[-0.28, +0.07]$
	Multilingual	0.473	0.455	-0.018	$[-0.08, +0.05]$
	Pooled	0.504	0.491	-0.014	$[-0.09, +0.07]$
AUROC within fold, size-weighted	Verified	—	—	-0.152	$[-1.00, +0.40]$
(3/9, 12/37, 15/46 folds usable;	Multilingual	—	—	-0.145	$[-0.38, +0.42]$
22/34, 31/77, 53/111 rows)	Pooled	—	—	-0.119	$[-0.31, +0.28]$
Fixed unit weights, no fitting	Verified	0.614	0.583	-0.030	$[-0.17, +0.25]$
($z(-c) + z(u)$, train-fold scaling)	Multilingual	0.561	0.610	$+0.048$	$[-0.07, +0.18]$
	Pooled	**0.559**	**0.571**	$\mathbf{+0.013}$	$[\mathbf{-0.07}, \mathbf{+0.12}]$

Table 3: The registered confirmatory endpoint under four estimators, repository-clustered bootstrap. Point estimates disagree in sign; every interval includes zero. The within-fold estimator drops repositories where no AUROC is defined and covers about half the rows, so its interval is wide; among the three full-coverage estimators the loosest pooled upper bound is +0.12. Raw AUROC($-c$) is 0.621 / 0.560 / 0.560.

a single window, and adds nothing to the ruler out of fold. We propose lines-added as the baseline any audit statistic must clear before it is worth a confirmatory run — and note that on SWE-bench Verified, beating it by 0.10 in a paired test has power of about 0.3 at any n, because twelve repositories cap the effective sample (§6).

5 Mechanism: stable critics, uninformative disagreement

Draft v0.1 of this paper read Fig. 8 — the same reward model scoring the prefix and the patch of one trajectory with correlation 0.03–0.18 — as evidence that these critics "do not hold a judgment" of code correctness. That reading could not distinguish critic noise from windows that legitimately contain different information, and we pre-registered a test to separate them (Amendment 4b, with the reading fixed in advance). The result withdraws the claim. Re-scoring the identical window after a reformatting that changes no information, every critic reproduces its ranking at $r = 0.98$–0.99; after a 128-token shift, at 0.86–0.95; the per-window panel u itself at 0.91 and 0.73. Adjacent windows with 25% overlap correlate at 0.20–0.56, first against last at -0.39 to 0.47. These critics are stable instruments responding to what is in front of them. The prefix score is a judgment of the problem; the patch score is a judgment of the patch; they differ because the inputs do.

The failure is therefore in the statistic, not the critics. A panel's disagreement is large when its members' rank-normalized scores spread out — which they do for reasons that have nothing to do with correctness: different critics are calibrated to different aspects of a window, and the spread is dominated by that between-critic heterogeneity rather than by any shared uncertainty about the solution. The signal the critics carry individually (0.71–0.75 at length) is averaged away, not amplified, by measuring how much they disagree. Windowed $u_{\max}$ recovers some of it, plausibly because the maximum over windows behaves partly like a max over the critics' own low scores; its exploratory single-critic analogues (Fig. 5, right) are consistent with that.

This explains the earlier false positive too. On HumanEval+ with an LLM-as-judge critic, we briefly observed $u = 0.69$ and a "Goodhart signature" of the critic below chance. Both were artifacts of a critic with three output values on three positives; with a real reward model selecting, both vanished. We

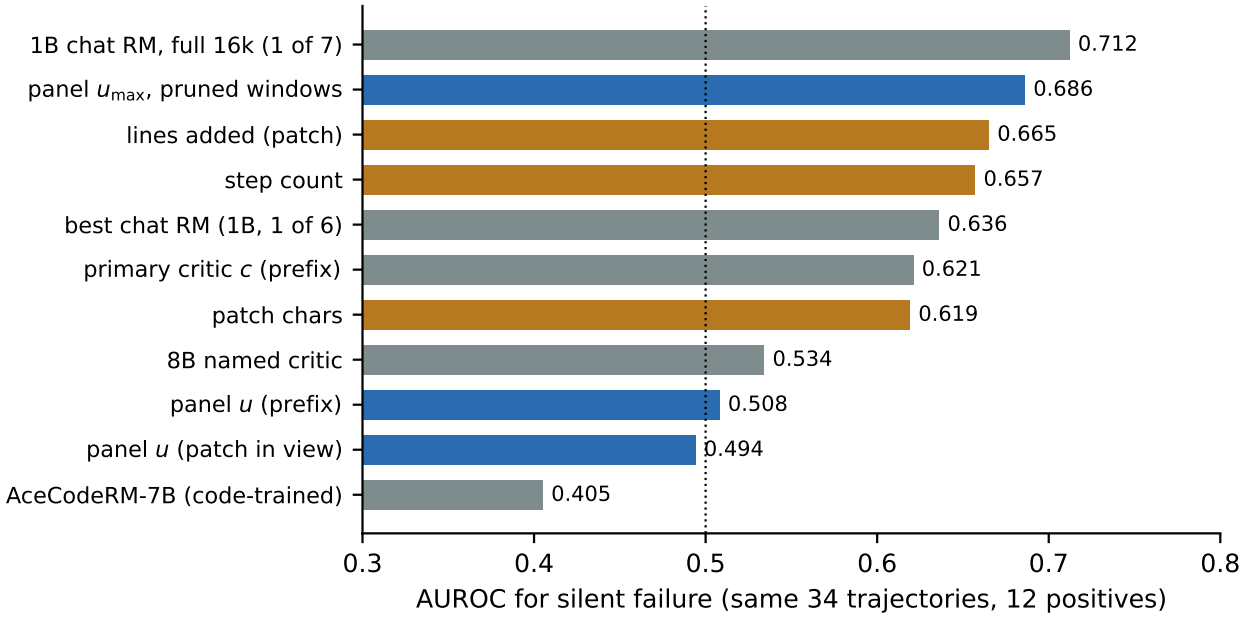


Figure 7: Same 34 trajectories, same 12 positives. Gold bars need no model. Exploratory (declared as such): the covariates were registered as controls, not baselines.

withdrew the claim in the record before running the SWE-bench pilot, as we withdraw the instability claim here.

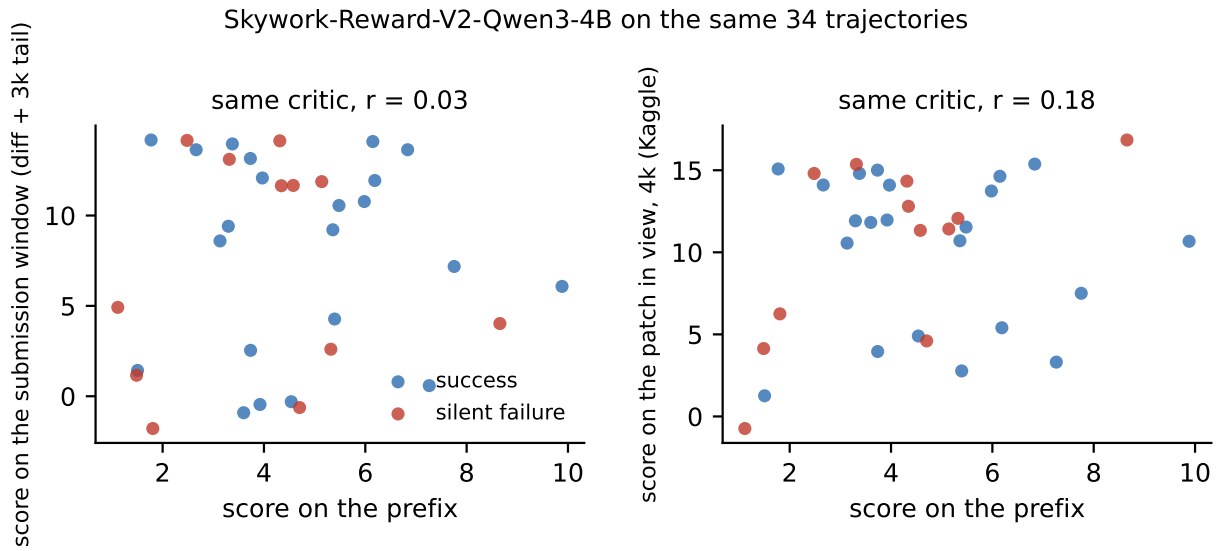


Figure 8: The primary selecting critic's score on the trajectory prefix (x) against its score on two other windows of the same trajectory (y). Same model, same trajectory, nearly unrelated numbers — because the windows contain different things (see Fig. 9).

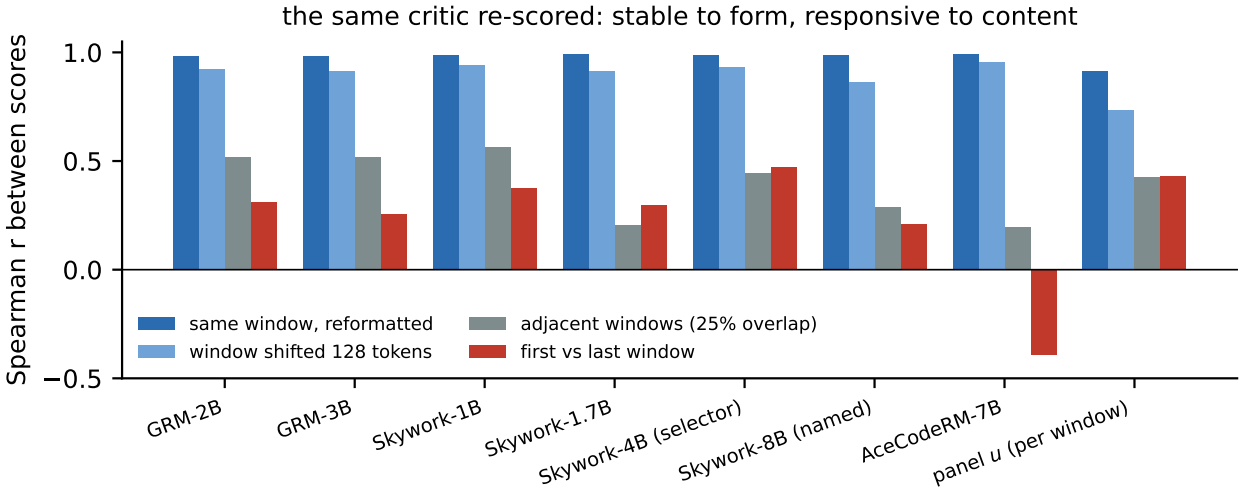


Figure 9: Amendment 4b, test–retest. Each critic re-scored every Amendment-3 window after a content-preserving reformatting (headers re-cased, pruning marker reworded, one neutral line prepended) and after a 128-token shift. Spearman r over 65–73 windows. The same critic on the same content is essentially deterministic; on adjacent or distant windows it is not.

6 Limitations and declared deviations

- **One agent, one selector family.** gpt-5-mini at 40 steps; a 4B chat reward model selects, so the selected population is that of a weak selector. Failure structure may differ for stronger agents or deployed-grade verifiers.
- **Selection did not concentrate failure here.** The unselected-candidate control (Amendment 4f) shows the committed candidate failing no more often than a passed-over one on either benchmark. The proposal's premise — that selection *enriches* the committed set for critic mistakes — is therefore not demonstrated for this selector; what is demonstrated is that selection removes the selector's own signal (0.81 → 0.61 on Verified). A stronger selector could change both numbers.
- **The selection-conditional regime is observed on one benchmark.** On Multilingual the selecting critic had no signal in either arm (0.43 unselected, 0.51 committed), so "residual value after the selector has consumed its signal" is measured only on Verified, where the selector's own AUROC fell from 0.81 to 0.61. The replication settles the null for disagreement; it does not by itself place the study in the regime the proposal named. A selector that ranks failures in the pool is the missing experiment.
- **Critic classes.** Six chat-preference RMs, two code-trained RMs and three LLM judges from one vendor; the 32B ran at 8k on most rows. Cross-vendor judges and trained trajectory verifiers were not run; the claim is scoped to off-the-shelf critics.
- **Gold is the harness verdict.** It accepts some incorrect patches and rejects some correct ones [40, 42, 53]; failure rates are lower bounds and every AUROC is attenuated by label noise.
- **Forty-seven positives.** Single-AUROC intervals are ±0.25 (Verified) and ±0.14 (Multilingual). Roughly twenty-five analyses were run on the twelve Verified positives without multiplicity correction; the replication is the correction, and of the pre-registered predictions that passed on Verified, none passed on Multilingual. The study supports failure to reject and an upper bound, not equivalence to chance; on the registered endpoint the bound is +0.04 pooled for the leave-one-repository-out logistic, +0.12 pooled across the three full-coverage estimators of it, and +0.25 on Verified alone (Table 3); on single-protocol comparisons it is +0.22, and the widest interval in the sweep reaches +0.45. We quote the loosest that applies to the population in question.
- **Repository clustering.** Verified is 500 instances over 12 repositories (django 46%); from six public submissions the ICC of resolution by repository is 0.033–0.053, an effective n of 64–95 of 500 (Fig. 2). Clustered bootstraps with nine clusters are anti-conservative; we report null-calibrated coverage in the record. The paired "beat the ruler by 0.10" test has power ≈0.3 on Verified at any n, which is why the replication changed repositories rather than instances.
- **The proposal's mechanism experiment (diversity vs. error correlation) was not run.** The m=3 family-disjoint robustness row is not a substitute.
- **Input length and windows.** The primary scored a 4k prefix; the six chat RMs were scored at a true 8k and 16k; `AceCodeRM-7B` fell back to 4k on 153/164 rows. Pruning left most trajectories at one to three windows; $u_{\max}$ is a maximum of noisy statistics and biased upward under the null.
- **Declared deviations.** 41 instances, not 50 (generation stalled on host disk; the missing nine were set by list order). Amendment 4a's registered difficulty control was a near-oracle (0.95) and was replaced by a labelled structure-only base. The replication's $15 generation cap was exceeded ($18.30; the account's rate limit killed 138 partial trajectories) because stopping at 420/456 would have dropped the harder instances first. 26% of replication candidates submitted no patch; 7 committed patches were ungradable and excluded.

7 Related work

Best-of-n, Goodhart and the winner's curse. Selecting the argmax of a learned proxy over-optimizes it [3, 11]; at inference time the effect appears as reward hacking that grows with n [4, 21, 47] and can be mitigated by pessimism [61]. The statistical form is the optimizer's curse [46]; selecting on a noisy score and then studying the selected set is selection bias in the structural sense [14]. Two recent notes make the evaluation consequence explicit: success counted before an answer is chosen is not deployed success [9], and best-of-n degrades calibration even under a perfect ranker [31]. Our estimand is this consequence for auditors.

Reward-model uncertainty and ensembles. Ensembles mitigate but do not eliminate over-optimization [6, 8]; uncertainty-aware reward models [30] and calibrated process reward models [37] change *selection*. We ask a different question — given the trajectory the selector already committed, does a disjoint panel's disagreement carry residual information — which none of these evaluate.

Correlated judges and panel effective size. Capable models make correlated errors [13, 22]; judge panels supply few effective votes [15, 23]; common-mode error is not identifiable from internal scores [2]; consensus can rival trained verifiers where errors are decorrelated [29, 48] and disagreement can be structured into a confidence estimator [17]. On code, verification helps only on margin-pivotal cases and a single round of debate erases the disagreement it needs [43, 44]. These predict our null; we supply an executed-gold instance on selected agent trajectories.

Process and outcome verifiers. Step-level supervision [26, 52], generative verifiers [64] and agent PRMs [54] score and select; they are not evaluated as residual auditors of a selected set.

SWE agents, verifiers and gold. SWE-bench and its Verified and Multilingual subsets [18, 35, 57], with mini-SWE-agent as the reference scaffold [59], are our testbed; SWE-smith, SWE-Gym and R2E-Gym supply training data and verifiers [16, 36, 58]; Agentless questions the need for agency at all [55]. Execution-free critics and reward models for SWE trajectories [45, 56] are evaluated on discrimination, calibration and test-time scaling over candidate pools; trained failure locators and prefix monitors [39, 51] and sequential-testing verifiers [41] target failure directly. None conditions on an upstream selector. The gold itself is imperfect: accepted patches can fail developer tests, test suites can be weak, and passing runs can be lucky [40, 42, 50, 53].

Agent failure and self-uncertainty. Agents declare success falsely [1], are overconfident about their own success [20], and their early uncertainty does not predict failure [25]; online auditors have been trained for multi-agent settings [63]. Semantic entropy and self-knowledge [10, 19] and selective classification [12] are the uncertainty tools these build on; long-context position effects [28] bear on what a critic sees in a window.

Reward models used. Skywork-Reward-V2 [27], GRM [60] and AceCodeRM [62], chosen by RewardBench standing [24, 32].

Methodology. Paired AUROC comparison for nested models should not use the DeLong test [7, 38]; clustered ROC methods [34] and decision curves [49] inform our analysis. The pre-registration trail follows the registered-report tradition [5, 33].

8 Conclusion

A third to a half of what a reward-model critic confidently committed on two SWE benchmarks was broken. The proposed safeguard — disagreement among a disjoint panel of off-the-shelf critics — did not yield a replicable failure-detection signal under any configuration we pre-registered and ran, across reward models and LLM judges alike, on fifty repositories. One configuration crossed its bar on the pilot: the maximum disagreement over pruned windows reached 0.686 on twelve positives, with a free permutation p of 0.04 that rose to 0.21 within repository. It did not survive replication. The critics are not noisy: they are stable, content-driven, and given the solution at length a frontier judge ranks silent failures at about two-thirds. What fails is the idea that their *disagreement* is where the missing information lives. The point estimates that suggested otherwise on twelve positives were settled the way they needed to be, on forty-one repositories rather than five hundred more trajectories, and they are at chance. Anyone building an escalation trigger for critic-selected agents should evaluate it where it will be used — on the selected population, against the selector's own score, with executed gold — and should expect off-the-shelf disagreement to add nothing detectable: on the registered endpoint, no estimator of it separates from zero, and the loosest bound we can defend on the pooled population is +0.12 AUROC over the selector's own score.

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A Chronology and cost

Stage	What	Cost
Freezes, defects, calibrated power	18 regression tests	$0
HumanEval+ pilots 1–3b (instrument validation)	40 tasks, $n \in \{4, 16\}$, 2 judges $\rightarrow$ 4 RMs $\rightarrow$ RM selection	$0
SWE-bench infrastructure without admin rights	Colima + Rosetta + swebench 5.x; three oracle controls	$0
SWE-bench pilot	164 gpt-5-mini candidates, 41 instances, official gold	$3.65
Amendment 1 (submission window)	local, 5 RMs	$0
Amendment 2 (Kaggle 2×T4)	7 critics, 164 candidates, two re-selections graded	$0
Amendments 2 (true 16k) and 3 (Algoverse L4)	7 critics $\times$ 164 candidates; 7 critics $\times$ 34 windowed rows	$0
Amendments 4a, 4b, 4d (local; L4 stage 5)	incremental-value model; test–retest 7 critics $\times$ 3 variants; 8k ladder point	$0
Amendment 4c (OpenAI API)	3 judges $\times$ 2 conditions $\times$ 34	$1.96
Modal (A100-80GB; CPU sandboxes)	AceCodeRM-32B $\times$ 164 + windows; official gold for 9 re-selected patches	≈$3
Amendment 4f control (Modal)	official gold for 144 unselected candidates, both benchmarks	≈$8
Amendment 4e replication (OpenAI + Modal)	456 trajectories, 114 instances, 41 repos; 7 critics at 4k/16k/windows; gold with 3 oracle pairs	$18.30 + ≈$8

B Operating points and alternative disagreement statistics

The escalation-trigger framing implies a review budget. Table 4 gives recall and precision when the top 5/10/20% of committed trajectories by each signal are flagged (positives: 12 of 34; 35 of 77). Table 5 replaces the registered u (sample SD of four rank-normalized scores) with other functions of the same panel scores; none is above 0.60.

C Reproducibility

Code, pre-registrations with amendments, every scored row, every critic's scores, the harness reports and the research log are in the repository (`UPDATES.md` has one entry per stage: done / learned / changed). All figures are generated by `paper/make_figures.py` from committed JSON.

Benchmark	Signal	recall at budget 5%	10%	20%	precision at budget 5%	10%	20%
Verified	panel u	0.08	0.08	0.25	0.50	0.33	0.43
	$-c$	0.17	0.17	0.33	1.00	0.67	0.57
	lines added	0.08	0.17	0.42	0.50	0.67	0.71
	gpt-5 judge	0.08	0.17	0.33	0.50	0.67	0.57
Multilingual	panel u	0.06	0.06	0.23	0.50	0.25	0.53
	$-c$	0.11	0.17	0.29	1.00	0.75	0.67
	lines added	0.03	0.09	0.17	0.25	0.38	0.40
	gpt-5 judge	0.09	0.14	0.23	0.75	0.63	0.53

Table 4: Review-budget operating points. Prevalence 0.35 and 0.45; a random flag has precision equal to prevalence.

	SD (registered)	range	mean abs. dev.	entropy	−mean rank	−min rank
Verified	0.508	0.559	0.481	0.500	0.466	0.483
Multilingual	0.565	0.587	0.552	0.393	0.601	0.581

Table 5: AUROC for silent failure of alternative statistics of the same four rank-normalized panel scores (exploratory).